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Claude Models, Prompting & Context Engineering - Test Complete!

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100%
Overall Score

10
Correct Answers

10:40
Time Taken

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Question Review

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1 A software company classifies 180,000 support tickets daily into 25 routing queues. The production SLA is 800 ms, and the minimum acceptable macro F1 score is 0.92. On a representative evaluation set, a faster model achieves 0.935 macro F1 at 420 ms and $0.003 per ticket; a more capable model achieves 0.947 at 1.1 seconds and $0.014 per ticket. Which model strategy should the architect recommend?

A. Use the more capable model because its higher F1 score outweighs the latency and cost differences.

B. Use the faster model because it meets the measured quality threshold, latency SLA, and operating constraints. **Correct**

C. Run both models for every ticket and accept a route only when their classifications agree.

D. Alternate between the models evenly so production traffic continues to exercise both capability levels.

2 Twelve business units use Claude to generate weekly operations reports. Each unit copied the original prompt and now maintains its own version. After six months, required risk disclosures are missing in four units, formatting differs across reports, and changes cannot be traced reliably. Units still need distinct terminology and metrics. Which prompt

architecture should the technical lead implement?

A. Select the best current unit prompt as the standard and prohibit business-unit-specific prompt changes.

B. Continue separate prompts but use a model to compare their wording and flag substantial differences weekly.

C. Version a shared core prompt and schema, with bounded unit-specific fields and tested override precedence. **Correct**

D. Combine every unit's instructions into one comprehensive prompt and let users specify the desired report type.

3 A migration assistant converts legacy batch jobs into a deployment pipeline. In testing, some outputs contain sound migration logic but invalid manifest structures; other outputs produce valid manifests that omit required rollback steps. The pipeline must consume artifacts automatically, but failed deployments can interrupt payroll processing. Which design BEST addresses both failure modes?

A. Use one prompt with stronger capitalization and repeated instructions requiring valid manifests and rollback steps.

B. Generate the complete artifact twice and deploy whichever output receives the higher score from another model.

C. Fine-tune output examples for manifest syntax, then allow the deployment pipeline to accept all generated artifacts.

D. Separate analysis and artifact generation, then apply schema and semantic validation with bounded repair or escalation. **Correct**

4 An enterprise procurement assistant supports negotiations that span several weeks. Its context currently contains every message, retrieved policy excerpt, and tool result. After long sessions, latency triples and the assistant sometimes follows superseded price targets. Current supplier status can

change daily, while legal requires citations for policy-based recommendations. Which context strategy is BEST?

A. Compact stale dialogue into structured state, refresh volatile supplier data, and retain authoritative evidence with provenance. **Correct**

B. Keep the full conversation but move the newest messages to the beginning so they receive greater attention.

C. Discard all earlier content after each session and ask users to restate the complete negotiation state.

D. Summarize the full conversation into narrative prose and remove all retrieved passages and citation metadata.

5 A pharmaceutical research assistant synthesizes internal studies, regulator notices, and licensed journals. The current system places up to 160 potentially relevant documents into each request. Evaluations show that the needed evidence is usually present, but answers sometimes favor repetitive older studies over recent regulator notices. The response SLA is 12 seconds, and every claim must remain traceable. Which change BEST addresses the failure?

A. Preserve all documents but repeat instructions that recent regulator notices must receive the highest priority.

B. Use a larger-context model and include additional studies so the model has more evidence for each synthesis.

C. Select evidence by relevance, authority, freshness, and diversity, preserve provenance, and expand sources progressively. **Correct**

D. Summarize every document with a fast model and provide only those summaries without links to original evidence.

6 A contract extraction service handles 70,000 requests daily. Every request uses the same 18,000-token policy guide and output schema, but cache hit

rates remain below 8%. Engineers construct prompts by interleaving customer metadata, the contract, policy sections, and schema fields in request-dependent order. Output quality is acceptable. Which change should the architect recommend FIRST?

A. Move all customer contracts into a shared cached prompt and append the policy guide to each request.

B. Create a versioned stable prefix for shared guidance and schemas, then append request-specific content consistently. **Correct**

C. Upgrade to a more capable model so each request requires fewer extraction instructions and schema examples.

D. Summarize the policy guide separately for every request before submitting the contract extraction prompt.

7 A global operations team is building an incident assistant for 240 services. Each service has runbooks, dashboards, diagnostic commands, and known-failure examples owned by a different team. Most incidents involve one or two services, and guidance changes weekly. Loading the entire catalog makes initial responses take 18 seconds against a 5-second target and occasionally mixes incompatible procedures. Which context design is BEST?

A. Load the complete runbook catalog once per session and instruct Claude to ignore services unrelated to the incident.

B. Fine-tune the weekly runbook catalog into the model so incident requests no longer need operational context.

C. Maintain separate full system prompts for all services and let operators choose the prompt before every incident.

D. Keep a small investigation framework, identify incident scope, and load versioned service modules on demand. **Correct**

8 A software company classifies 600,000 support tickets per day before

routing them to specialist queues. Testing on representative production tickets shows that a smaller Claude model achieves 97.2% routing accuracy at 350 ms and $0.004 per ticket. A more capable model achieves 97.8% at 1.1 seconds and $0.019 per ticket. The approved targets are at least 97% accuracy, less than 800 ms, and no more than $3,000 per day. Misrouted tickets are reversible through an existing triage queue. Which model strategy should the architect recommend?

A. Use the more capable model because its higher aggregate accuracy minimizes all downstream operational risk.

B. Use the smaller model because it meets validated quality and latency targets within the daily budget. **Correct**

C. Send each ticket to both models and accept the classification when their outputs agree.

D. Use the smaller model initially, then randomly reprocess half of all tickets with the more capable model.

9 An operations assistant can inspect Kubernetes workloads and restart services. Its system prompt says never to modify production without approval, but a retrieved incident note containing malicious instructions caused the assistant to call a production restart tool. Restarting a service can interrupt thousands of active sessions. Engineers need autonomous diagnostics, but production changes require an on-call engineer's approval. Which change most directly reduces the risk?

A. Require an external approval token at the restart tool boundary and restrict the assistant's default credentials to read-only diagnostics. **Correct**

B. Expand the system prompt with examples of malicious incident notes and repeat the production-change prohibition.

C. Record all prompts and tool calls in a security dashboard and alert when a production restart occurs.

D. Use a more capable model to identify conflicting instructions before it selects a production tool.

10 A financial services company is building a Claude-based analysis platform for twelve business units. Each unit needs different analysis instructions and examples, but compliance rules and required disclosure fields are common. Today, teams edit copies of one large prompt, making policy changes difficult to audit and causing output-schema regressions. Source documents may contain untrusted instructions. Which prompt architecture should the platform team adopt?

A. Maintain one centrally owned prompt containing every unit's policies, examples, source documents, and conditional formatting rules.

B. Let each unit own its complete prompt, then run a language-model review to detect policy deviations before release.

C. Store compliance text in retrieved documents, allow each unit to write its own prompt, and reject outputs missing disclosure phrases.

D. Version stable system controls centrally, expose bounded task templates, delimit source

data, and validate required output fields externally. **Correct**

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